Thomas De Min

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Experience

INRIA | Research Scientist Intern
February 2023 - May 2023 | Grenoble, France

- Developed a novel Class-Incremental Learning approach that exploits Parameter-Efficient Fine-Tuning and pre-trained Vision Transformers.
- The method achieved a new state-of-the-art on well-known benchmark datasets while being more efficient than previous approaches in the number of parameters trained and computational requirements.
- The work was published at the ICCV workshop "Visual Continual Learning."

UNIVERSITY OF TRENTO | University Tutor
September 2022 - June 2023 | Trento, Italy

- Tutor for the Programming 1 Bachelor's course: Designed exercises, and exam simulations and evaluated them to help students understand the course topics.
- Tutor for the Deep Learning Master's course. Evaluated Visual Grounding projects and assessed student knowledge via oral examinations.

BLUETENSOR S.R.L. | A.I. Engineer Intern
May 2021 - October 2021 | Trento, Italy

- Developed Computer Vision solutions for panoptic segmentation, object recognition, and face re-identification.
- Engineered an approach to cluster people's behavior on a website using common Machine-learning approaches and the BERT Language Model.
- Designed a multi-camera tracking infrastructure for recognizing targets across different environments.

Projects

HUMAN-MACHINE DIALOGUE | RASA
2023

- Designed and trained a Conversational A.I. for managing Cinema Reservations. The project was evaluated using an Alexa Dot.

DOMAIN ADAPTATION | PyTorch
2022

- Implemented and improved the Contrastive Adaptation Network (Kang et. al.) in order to cope with domain shift. Training and Testing were performed on a subset of the Adaptiope Dataset.

SENTIMENT ANALYSIS | PyTorch
2022

- Trained and evaluated different Machine Learning models (Naïve Bayes, GRU with Local Attention, and BERT) for polarity classification of movie reviews and subjectivity detection.

ASSET TRADING | PyTorch & DEAP
2022

- Trained and compared Actor-Critic and Genetic Programming in trading assets in the stock market.

Publications

See the full list of publications I contributed to on my scholar profile.

Reviewer

CVPR 2024 • RA-L • ICPR 2024

Education

UNIVERSITY OF TRENTO
Ph.D. student in Information Engineering and Computer Science
Nov. 2023 - Nov. 2026 | Trento, Italy

UNIVERSITY OF TRENTO
Master's Degree in Artificial Intelligence Systems
Sep. 2021 - July 2023 | Trento, Italy
Grade 110/110 Cum Laude

Skills

PROGRAMMING

Proficient:
Python

Experienced:
C • C++ • Java • LaTeX • Shell • SQL • JSON

Familiar:
HTML • CSS • Javascript

LIBRARIES/FRAMEWORKS

PyTorch • Scikit-learn • Numpy • Eigen (C++)

TOOLS/PLATFORMS

Git • WandB • Docker • D-Wave QPU • Colab